Michael D. Sudano

457 Everett Avenue, Palo Alto, CA 94301
(917) 923-3664
msudano@gmail.com
www.linkedin.com/in/msudano
www.mecha-sensei.com

EXPERIENCE

KAWASAKI ROBOTICS (USA), INC. [San Jose, CA]
Robotics Engineer (Jan 2011 - May 2014)

- DFM, DFA of links and end effectors of silicon wafer-handling robots
- GD&T using AutoCAD for space optimization and key features, datum surfaces
- Unigraphics NX 3D modeling of robotics components, assemblies and fully integrated systems
- ANSYS FEA analysis and redesign of existing robotic components for stress, strain and vibration
- SCARA and 6-axis robot programming, both by command-line and teach pendant
- Hysteresis, repeatability testing, assembly, installation, calibration and repair in Class 1 cleanroom
- Extensive customer and internal interaction in both Japanese and English

HONEYBEE ROBOTICS SPACECRAFT MECHANISMS CORPORATION [New York, NY]
Project Engineer (Dec 2006 – Nov 2010)

- AutoCAD 2D design, Autodesk Inventor 3D modeling of electronics enclosures and control panels
- Lathe, mill machining of aluminum, steel and plastic parts to sub-mil tolerances
- ESD-controlled electromechanical assembly and harness fabrication
- Linux-based C++ robot programming, including multi-threading
- Testing of the next-generation NASA Mars Exploration Rover Rock Abrasion Tool (RAT)

NASA AMES RESEARCH CENTER – INTELLIGENT ROBOTICS GROUP [Mountain View, CA]
Robotics Academy Program Group Leader (Jun 2007 – Aug 2007)

- Design and construction lead of a non-prehensile robotic platform for lunar excavation
- SolidWorks modeling of entire robotic assembly, including electronics components and enclosures
- LabView creation of a touchscreen GUI for robot control, including video via webcam
- Electromechanical assembly, debug and testing of the robot in a simulated lunar environment
- Mass, structural strength and power calculations for redesign of lunar orbiter subsystems

EDUCATION

THE COOPER UNION FOR THE ADVANCEMENT OF SCIENCE AND ART [New York, NY]
Master of Mechanical Engineering May 2010 – GPA 3.8, summa cum laude, full scholarship
Bachelor of Mechanical Engineering May 2005 – GPA 3.8, summa cum laude, full scholarship

PUBLISHED WORK

Navigating through the Magellan Pro Robot — Master's Thesis, The Cooper Union, 2010
Geotechnical Property Tool on NASA Ames K-10 Rover — LEAG-ICEUM-SRR, 2008
A Low-Complexity Navigation Algorithm for a Scalable Autonomous Firefighting Vehicle — IEEE, 2007

KEY SKILLS

Industrial Robotics, Mechatronics, Mobile Autonomous Robotics, Semiconductors
Unigraphics NX, SolidWorks, Autodesk Inventor, AutoCAD, DFM, DFA, GD&T, FEA, ANSYS, COSMOS
System Integration, Field Application, Fabrication, 3D Printing, Japanese